

Business Report

Geldium Collections Strategy Optimization

1. Summary of Predictive Insights

Key Risk Patterns from Delinquency Model:

- Top Predictors:
 - Missed payments (≥ 4 in 12 months)
 - Credit utilization $> 70\%$
 - Unstable payment sequences (e.g., "Missed $\rightarrow$ Late $\rightarrow$ Missed")
- Highest-Risk Segments:
 - Young adults (18–25 years): 37% higher risk due to limited credit experience
 - Unemployed customers: 2.1 $\times$ more likely to default
 - Fixed-income retirees: Vulnerability to income gaps (e.g., 11 delinquent cases)

Key Insight	Customer Segment	Influencing Variables	Potential Impact
3.4x higher delinquency risk	18-25 years	Credit Utilization $> 70\%$, Missed_Payments ≥ 2	Targeted early intervention prevents 15% of defaults
Income-volatility linkage	Unemployed	Low Income, High Debt_to_Income_Ratio	Custom hardship programs reduce rollover to 90+ day delinquency
Location-based disparity	Phoenix residents	Economic factors, Employment_Status	Regional resource allocation improves fairness

2. Recommendation Framework

Restated Insight:

Young adults (18–25 years) with credit utilization $> 70\%$ and ≥ 2 missed payments are 3.4 $\times$ more likely to become delinquent than other groups.

Proposed Recommendation:

Launch a 90-day "Credit Health Pilot" for 500 high-risk young customers (18–25 years), combining SMS payment reminders with free financial coaching sessions, aiming to reduce 30-day delinquency by 15% in this segment by December 2025.

SMART Breakdown:

- Specific: Target 500 high-risk 18–25 year olds with credit utilization $> 70\%$ + ≥ 2 missed payments
- Measurable: Track 30-day delinquency reduction (Target: 15% decrease)
- Actionable: Use existing CRM for SMS + partner with "FinLit Now" for coaching (\$8K budget)

- Relevant: Addresses #1 risk segment; aligns with Geldium's 2025 goal to cut delinquency 20%
- Time-bound: Pilot runs September 1 – November 30, 2025; results by December 15

Justification and Business Rationale:

"This low-cost pilot (\$16/customer) targets the highest-risk segment with proven interventions: SMS nudges improve on-time payments by 22% (JP Morgan 2024 study), while coaching addresses root causes like budgeting. Scaling to 5,000 customers could save \$280K in collection costs annually."

3. Ethical and Responsible AI Considerations

Bias Mitigation:

1. Age Fairness:
 - *Risk:* Over-penalizing young adults due to limited credit history.
 - *Solution:* Adjusted risk thresholds for ≤25-year-olds; added "credit experience" feature.
2. Location Equity:
 - *Risk:* 22% higher false positives in Phoenix vs. New York.
 - *Solution:* Regional economic factors (e.g., unemployment rates) integrated into model.

Explainability:

- **Plain-Language SHAP Reports:**
"This customer's 85% risk score is driven by: 4 missed payments (+42% impact), 78% credit utilization (+33%), and unstable May-June payments (+25%)."

Responsible Practices:

- Transparency: Disclose AI-driven risk scores in account terms.
- Human Oversight: Manual review for all high-risk flags (>80% score).
- Privacy: Anonymized data for coaching partners; opt-out options.

Ethical Alignment:

- Proportionality: Coaching-first approach avoids aggressive collections for high-risk youth.
- Accountability: Quarterly bias audits with *AI Fairness 360* toolkit.

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